

24 The Origin of Species

KEY CONCEPTS

- 24.1** The biological species concept emphasizes reproductive isolation *p. 507*
- 24.2** Speciation can take place with or without geographic separation *p. 511*
- 24.3** Hybrid zones reveal factors that cause reproductive isolation *p. 516*
- 24.4** Speciation can occur rapidly or slowly and can result from changes in few or many genes *p. 520*

Study Tip

Make a table: Some processes that can lead to speciation only occur in allopatric (geographically separated) populations, while others also occur in sympatric (geographically overlapping) populations. To help you keep track of these processes and the geographic conditions under which they can occur, fill in the table as you read the chapter.

Process (include page # or figure #)	Can occur in allopatric populations (yes/no)?	Can occur in sympatric populations (yes/no)?
Sexual selection (Figure 24.12)	Yes	Yes

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For Students (in eText and Study Area)

- Get Ready for Chapter 24
- Figure 24.10 Walkthrough: One Mechanism for Allopolyploid Speciation in Plants

For Instructors to Assign (in Item Library)

- Interpreting Data: Evolution of Reproductive Barriers in Lab Populations of Fruit Flies
- Problem-Solving Exercise: Is Hybridization Promoting Insecticide Resistance?



Figure 24.1 This flightless cormorant is one of many species in the Galápagos Islands that are found nowhere else in the world. How did a bird that cannot fly come to live in this isolated place? When Darwin visited the Galápagos, he also was intrigued by its unique species, and later concluded that they must have originated in the islands from ancestors that had traveled from South America.

How do new species originate from existing species?

Over time, populations of a single species connected by gene flow can diverge genetically, giving rise to a new species:

